



Merge text strings to gain insights

Great to see you back. In this video, we'll build on what we've learned about CONCATENATE and IMPORTRANGE by exploring a new SQL query: CONCAT. You might remember that CONCATENATE is a function that joins together two or more text strings. As a quick reminder, a text string is a group of characters within a cell most often composed of letters. You've seen how that works within a single spreadsheet. But there's a similar function in SQL that allows you to join multiple text strings from multiple sources, CONCAT. Let's use CONCAT to combine strings from multiple tables to create new strings.

For this example, we'll use open data from Citi Bike, which is a public bicycle sharing system in New York. As you've learned earlier, open data initiatives have created a ton of data for analysts to use. Openness or open data is free access, usage, and sharing of data. It's a great resource if you want to practice or experiment with the data analysis tools you've been learning here. You have open access to the New York city bike-sharing data, which has information about the use of shared bikes across the city. Now we can use CONCAT to pull and concatenate data from different columns stored here. The first thing we need to do is figure out which columns we need.

That way we can tell SQL where the strings we want are. For example, the bike-sharing company has two different kinds of customers; one-time paying customers and

subscribers. Let's say we want to find out what routes are most popular with different user types. To do that, we need to create strings of recognizable route names that we can count and sort. We know that the information we need is in the stations and trips table. We'll start building our query from there. First, we'll input `SELECT` user type to let SQL know that we want the user type as a column.

Then we'll use `CONCAT` to combine the names of the beginning and ending stations for each trip in a new column. This will create one column based on the routes people take. We also need to input a title for this new column. We'll type in, `AS route`, to name the route column using those beginning and ending station names we combined with `CONCAT`. This will make these route names easy for us to read and understand. After that, we want SQL to count the number of trips. So we'll input `COUNT` to do that.

We can use an asterisk to tell it to count up the number of rows in the data we're selecting. In this case, each row represents a trip, which is why we can just count all of the rows we've selected. We'll name this output as `num_trips`. Now let's also get the average trip duration for each route. In this case, we don't need the exact average, so we can use the `ROUND` function to round up. We'll put that first and then in the parentheses use `average` to get the average trip duration. We'll also want this data to be in integer form for this calculation, so we'll input `cast as int 64`.

Big query stores numbers in a 64-bit memory system, which is why there's a 64 after integer in this case. Next, we'll divide it by the number of rows and tell it how far we want it to round, two decimal places. We'll name this output as `duration`. We'll need to tell SQL where this information is stored. We'll use `FROM` and the location we're pulling it from. Since we're using `COUNT` and `AVERAGE` functions in our select clause, we

have to use GROUP BY to group together summary rows. Let's group by the start station, the end station, and the user type for this query.

Finally, we'll use ORDER BY to tell it how we want to organize this data. For this, we want to figure out the most common trips so we can input the number of trips column and use DESC to put it in descending order. Finally, we only want the top 10, so let's add LIMIT 10. Now thanks to CONCAT, we can easily read these route names and trace them back to real places. We can see which kinds of customers are taking which routes, which can help the bike-sharing company understand their user base in different parts of the city and where to keep more bikes for people to rent. Being able to combine multiple pieces of data can give you new ways to organize and analyze data. There's a lot of different tools to help you do that.

Now you've seen CONCAT in action, and later you will come across another similar query, JOIN. But up next, we'll talk more about working with strings. See you soon.